



High correlated variables creator machine: Prediction of the compressive strength of concrete

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ABSTRACT

In this paper, we introduce a novel hybrid model for predicting the compressive strength of concrete using Ultrasonic Pulse Velocity (UPV) and Rebound Number (RN). First, we collect 516 datasets from 8 studies of UPV and Rebound Hammer (RH) tests. Then, we propose the High Correlated Variables Creator Machine (HCVCVM) to create the new variables that have a better correlation with the output in order to improve the prediction models. Three single models, including a Step-By-Step Regression (SBSR), Gene Expression Programming (GEP), an Adaptive Neuro-Fuzzy Inference System (ANFIS) as well as three hybrid models, i.e. HCVCVM-SBSR, HCVCVM-GEP, and HCVCVM-ANFIS are employed to predict the compressive strength of concrete. The statistical parameters and error terms such as the coefficient of determination, the Root Mean Square Error (RMSE), Normalized Mean Square Error (NMSE), fractional bias, the maximum positive and negative errors, and the Mean Absolute Percentage Error (MAPE) are computed to evaluate the models. The results show that HCVCVM-ANFIS can predict the compressive strength of concrete better than all other models. HCVCVM improves the accuracy of ANFIS by 5% in the coefficient of determination, 10% in the RMSE, 3% in the NMSE, 20% in MAPE, and 7% in the maximum negative error.

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1. Introduction

Thanks to its low cost, concrete is among the most widely used building materials in the world. Estimating the properties of hardened concrete has been the topic of numerous studies [1]. The compressive strength is a key property of concrete and directly related to the safety of the structure. Evaluating the Concrete Compressive Strength (CCS) is a common requirement in quality control and performance determination of existing structures during their entire life cycle. Strength test results also may be a necessity after a structural failure, such as environmental degradation or fire damage [2]. Fig. 1.

Methods to estimate the mechanical properties of concrete can be classified into Destructive Tests (DTs) and Non-Destructive Tests (NDTs). Usually, cubic or cylinder specimen are prepared and cured to measure the compressive strength of fresh concrete

in accordance with procedures described in the American Society for Testing and Materials (ASTM) standard test methods [3,4]. Different concrete mixtures lead to different properties, and the reproducibility of the test results is of major importance [5–8]. Although DTs, such as core tests, are commonly used, they have several drawbacks. They are expensive and time-consuming and have 'locally destructive effects' on the structures. They are also the only representative for small parts of structures, and conducting these tests is sometimes difficult due to access to the coring machine [9].

NDTs are widely used to evaluate the fundamental characteristics of concrete in structures, such as compressive strength. NDTs do not cause any damage to the structure and do not affect the structural performance [10]. Many different types of NDTs have been developed in order to measure the compressive strength of concrete, among them the Ultrasonic Pulse Velocity (UPV) and Rebound Hammer (RH) tests. These methods are simple yet effective and can be used for quality control of concrete in structures or

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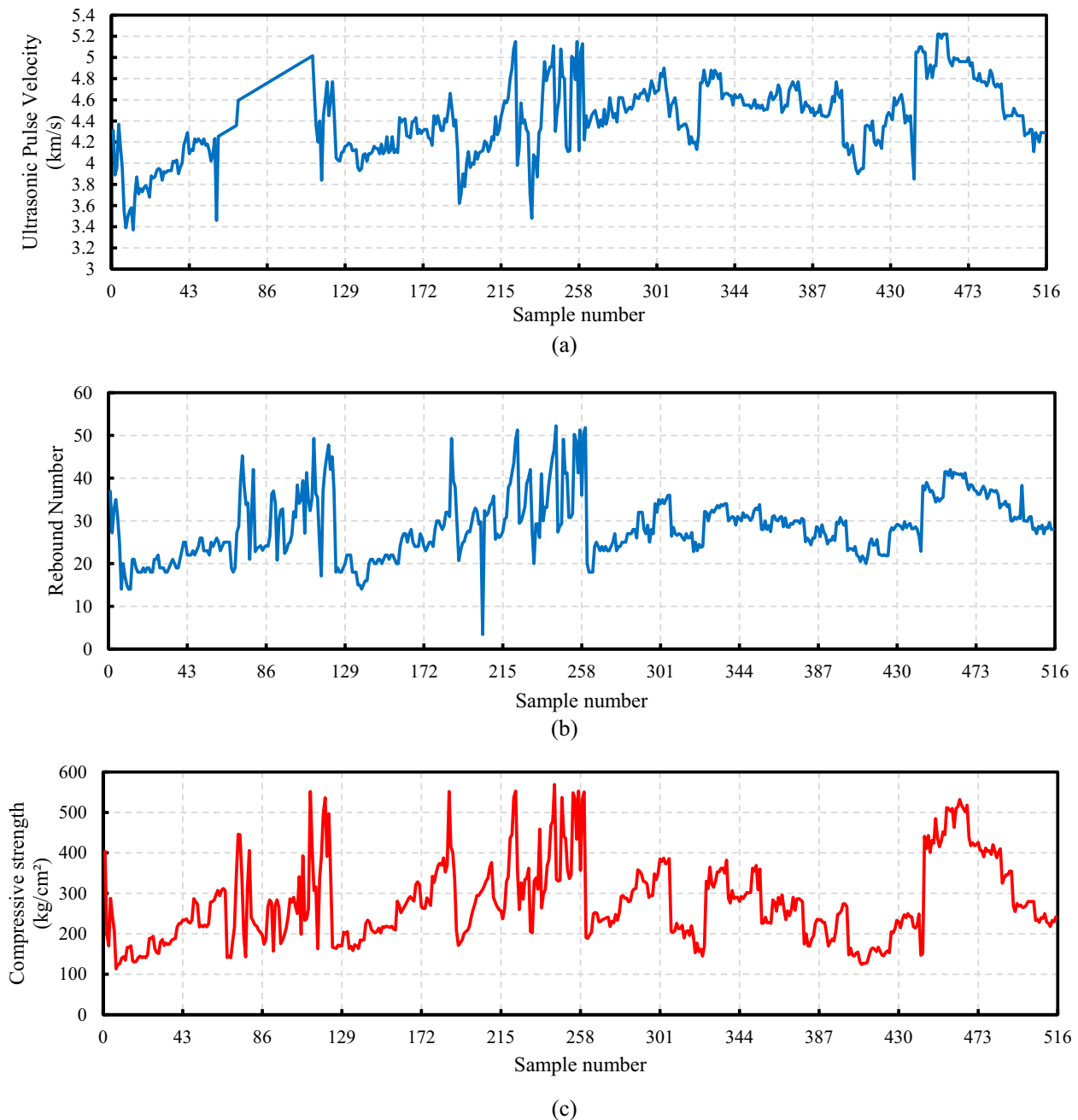


Fig. 1. The values of input variables and output. (a) UPV. (b) RN. (c) Compressive strength of concrete.

confirm the uniformity of the material from one part to another [11–13].

UPV is a common and fast test without any damage to concrete. It was presented and proposed by ASTM C 597 and BS 1881: Part 203 [4,14]. This test is one of the suitable approaches to check the uniformity of concrete quickly, and it can be applied in various parts of Reinforced Concrete (RC) elements. This test can determine the properties of used concrete in structures and also specify the presence of voids and internal cracks [15]. The results of this test depend on several factors, including the percentage of components of the mix proportion, properties of aggregates, steel bars, cracks, voids, and fibers. On the other hand, some studies report high uncertainties of the results of UPV to estimate the compressive strength of concrete [16,17].

The RH test is one of the most used non-destructive tests because of its simplicity. This test was presented by ASTM C 805 and BS 1881: Part 202 [14,18]. In this test method, a spring-loaded mass, which has a fixed amount of energy, is released. The traveled distance by this mass is defined as a percentage of the initial spring extension. This value is defined through a Rebound Number (RN). The determined RNs depend on the percentage of components of mix proportions, surface texture, surface wetness of the concrete, and properties of the aggregate, although the RN is the reflection of the concrete surface [11,17,19].

Sometimes, a combination of UPV and RH methods is used for better prediction of concrete compressive strength, which is also known as the SonReb method. UPV and RH tests are both sensitive to the variations in the concrete properties, especially in opposite

directions. UPV tests provide information about the interior concrete properties, whereas the RH test results usually predict concrete strength near the surface. Consequently, using the combination of two methods for the prediction can lead to more accurate and reliable results [17,20].

2. Theoretical background

Introducing precise and reliable models for predicting the CCS would lead to saving both costs and time. In this regard, many researchers have used several models to estimate the compressive strength of concrete [21–26]. Based on the results of the previous studies, computational modeling, Artificial Intelligence (AI) and parametric multi-variable regression models are three groups of techniques that are commonly used to predict concrete strength using the UPV and RH outputs. Computational modeling is based on the mathematical modeling of complex physical phenomena and often encounter some difficulties due to the complexity of the models or the time consuming numerical computation [19]. In this regard, several equations were established by researchers for predicting concrete compressive strength; however, most of them require previous knowledge of concrete ingredients to achieve reliable results. Among them, Rashid and Waqas [11] established a relationship between UPV with compressive strength and RN with compressive strength. Ahmadi-Nedushan proposed an optimized instance-based learning approach for prediction of the compressive strength of high-performance concrete based on mixed data, such as water to binder ratio, water content, superplasticizer content, fly ash content [25]. In recent years, AI methods and Machine Learning (ML) techniques have been used frequently to predict the CCS without knowing the theoretical relationships between the inputs and the output. Several researchers have employed AI methods to predict the mechanical properties of different types of concrete using detail of mix proportion and UPV [22,22,27,28]. Tenza-Abril et al. used an Artificial Neural Network (ANN) for the prediction and sensitivity analysis of compressive strength in segregated lightweight concrete using UPV [29]. Ashrafian [30] utilized heuristic regression methods as a hybrid method in the prediction of both strength and UPV for fiber reinforced concrete. Adaptive Neuro-Fuzzy Inference System (ANFIS) is another strong candidate for predicting the performance of concrete. For example, the properties of rubberized concrete, compressive strength, and splitting tensile strength of concrete containing silica fume and recycled aggregate at 3, 7, 14, 28, and 90 days, and compressive strength of no-slump concrete were predicted using ANFIS [21,31,32]. Cevik [33] employed ANFIS to predict the increased strength of Fiber-Reinforced Polymer (FRP) confined concrete. Several researchers used several methods, including ANFIS, ANN, and group method, to estimate the compressive strength of concrete based on the results of core tests. The used variables include length-to-diameter ratio, core diameter, aggregate size, and concrete age [34].

Parametric multi-variable regression models are another group of methods to predict CCS. Specifically, the confidence interval of a prediction calculated by a parametric regression model reveals the accuracy of the estimation, and the statistics of the model parameters can indicate how each variable in the model influences the prediction [35]. Kewalramani [16], used a regression analysis to predict the compressive strength of concrete based on acoustic characteristics like UPV and damping constant.

The main target of this study is a comparison of the accuracy and performance of two new proposed models and two available models in predicting the compressive strength of concrete using RN and UPV. First, the data is collected from 8 previous studies. Secondly, the Step-By-Step Regression (SBSR), Gene Expression Programming (GEP) and ANFIS are used as an individual model

to predict the compressive strength. Thirdly, High Correlated Variable Creator Machine (HCVCVM) is used to generate the new variables, which can improve the accuracy and performance of the models. HCVCVM is also introduced and utilized for the first time. Using the combination of HCVCVM and the used models in previous steps, three hybrid models, including HCVCVM-SBSR, HCVCVM-GEP, and HCVCVM-ANFIS, are created and used to predict the compressive strength. The statistical parameters and error terms, including the coefficient of determination, Root Mean Square Error (RMSE), Normalized Square Error (NMSE), fractional bias, the maximum positive and negative errors, and Mean Absolute Percentage Error (MAPE) are used to evaluate the models.

3. Materials and methods

The main target of this study is related to the comparison accuracy and performance of two unknown models and two known models in predicting the compressive strength of concrete using RN and UPV. First, the values of RN, UPV, and compressive strength of concrete from 8 previous studies are collected. Then, unknown regression models, namely SBSR, GEP, and ANFIS, are utilized to predict the compressive strength of concrete using the initial input variables, including the values of RN and UPV. Subsequently, HCVCVM is used to propose several new variables. The proposed new variables by HCVCVM are imported as input variables in SBSR, GEP, and ANFIS; therefore, HCVCVM-SBSR, HCVCVM-GEP, and HCVCVM-ANFIS can be regarded as hybrid models. The results of single models are compared with the results of the hybrid models. 70% of the dataset, which is selected to train models randomly, is the same for all models. After training the models, the remaining dataset is predicted using the trained models.

3.1. Data

To build up a precise prediction model, a large number of experimental data on normal-weight concrete compressive strength, UPV test results, and rebound number from rebound hammer tests is required. We have collected 516 data points of the compressive strength of concrete, which were tested at 3, 7, 14, 28, 56, and 90 days [9,10,13,15,17,36–38].

UPV and RN are considered as inputs of the models, and the compressive strength of concrete is selected as the output. Table 1 shows the variation range, average, median, and standard deviation of each variable, while the histogram of the involved variables can be found in Fig. 2.

Table 2 shows the coefficient of determination (R^2) between the input variables and also the coefficient of determination between the inputs and output. The coefficient of determination is a statistical measurement that examines how differences in one variable can be explained by the difference in a second variable [26,39]. In other words, this coefficient assesses how strong the linear relationship is between two parameters. The optimal value of the coefficient of determination between an input and the output is 1. Several input variables might have the same coefficient of determination with the output. In these cases, if the coefficient of determination between these input variables is close to 1, it means that the input variables represent the same properties and data. The results show that the input variables are independent, and RN has a better correlation with the compressive strength of concrete.

3.2. Methods

In this study, three single algorithms and three hybrid algorithms are used to predict the compressive strength of concrete using RN and PUV. The single algorithms include SBSR, GEP, and

Table 1
Statistical parameters of input variables and output.

Variable	Unit	min	max	average	SD	median	Type
UPV	Km/S	3.37	5.22	4.44	0.35	4.45	Input
RN	–	3	52	29	7.32	28	Input
CCS	Kg/Cm ²	113.33	569.21	276.92	98.60	261.97	Output

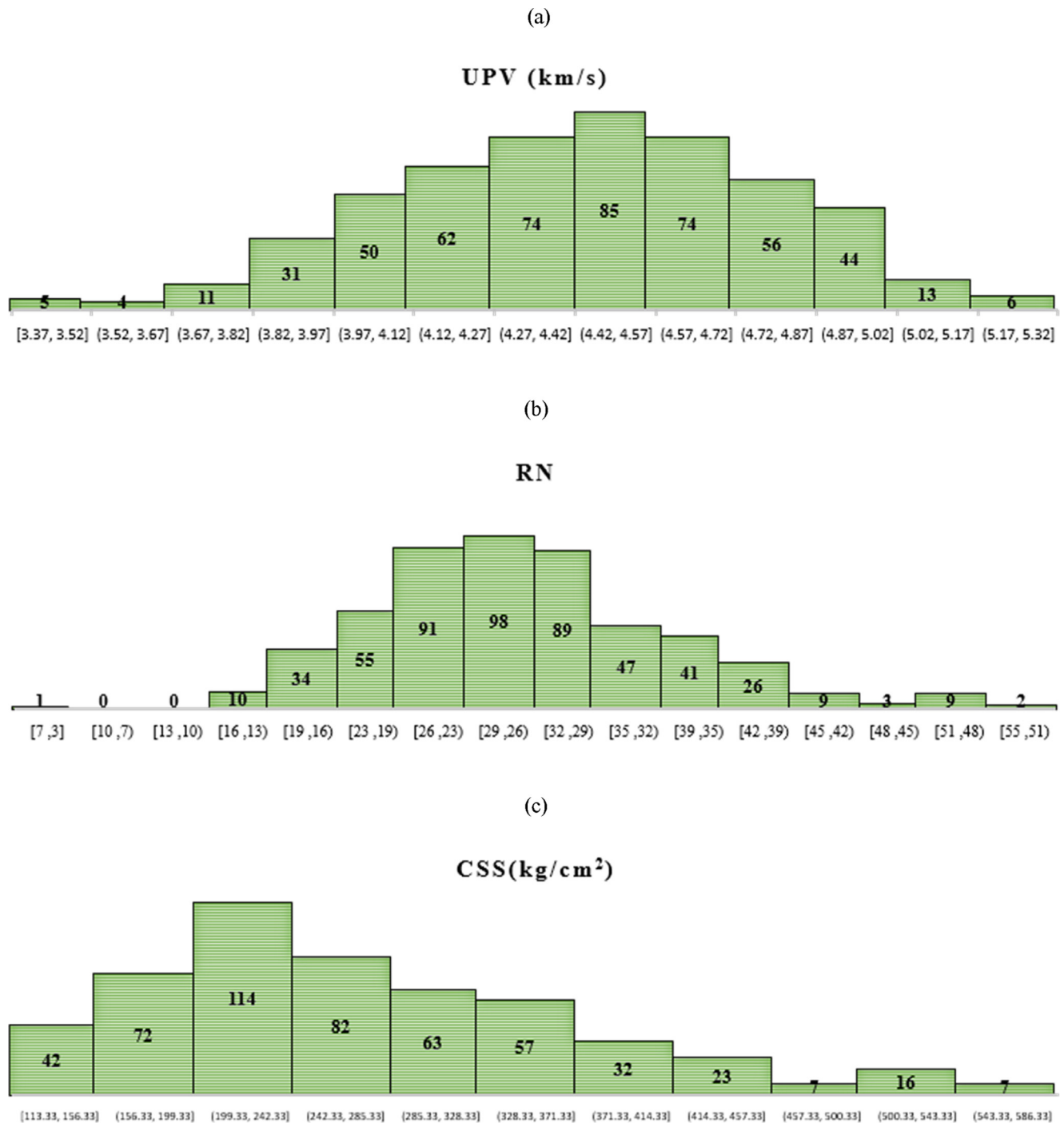


Fig. 2. The histogram of input variables and output. (a) UPV. (b) RN. (c) CSS.

ANFIS. The hybrid models are created by combining HCVC and each single algorithm. 70% of the dataset is selected randomly to train the models. Then, they are used to predict the remaining dataset.

3.2.1. Step-By-Step regression (SBSR)

SBSR is a new regression model, which focuses on the coefficient of determination between the variables and output to improve the accuracy of the obtained model. Fig. 3 shows an SBSR

Table 2

The coefficient of determination between variables and output.

Coefficient of determination (R^2)	
Between compressive strength and UPV	0.443
Between compressive strength and RN	0.758
Between RN and UPV	0.513

model for three variables. In the first step, the coefficient of determination between each input variable and output is calculated. Then, the variables are sorted based on the obtained values of the coefficient of determination. The variable with the highest coefficient of determination is imported in cell 1, and the variable with the smallest coefficient of determination is placed in the last cell (see the second step in Fig. 3). In the second step, linear regression is used to predict the output in each cell. As shown in Fig. 3 and equations 1–3, the outputs of cells 1 and 2 are used as input in cells 2 and 3, respectively. Then, the predicted outputs of all cells in the second step are imported as new variables in the final cell, which is located in the third step. These variables are used in the final cell to predict the output. In all cells, the constant parameters of the regression model are obtained using the mean square error (MSE) technique. Equations (1)–(3) show the regression model in the first cell, second cell, and third cell, respectively. Equation (4) represents the regression model in the final cell, which is located in the final step.

$$Y = b_1 + b_2 \times X_1 \quad (1)$$

$$Y = b_3 + b_4 \times X_2 + b_5 \times F_1 \quad (2)$$

$$Y = b_6 + b_7 \times X_3 + b_8 \times F_2 \quad (3)$$

$$Y = b_9 + b_{10} \times F_1 + b_{11} \times F_2 + b_{12} \times F_3 \quad (4)$$

where X_1 , X_2 , and X_3 refer to the sorted input variables based on the coefficient of determination with the output; F_1 , F_2 , and F_3 indicate the outputs of Cells 1, 2, and 3, respectively while b_1 , b_2 , ..., b_{12} denote the constant variable of the regression model, i.e.

$$b_i = \left(\frac{X^T}{X} \right)^{-1} \left(\frac{X^T}{Y} \right) \quad (5)$$

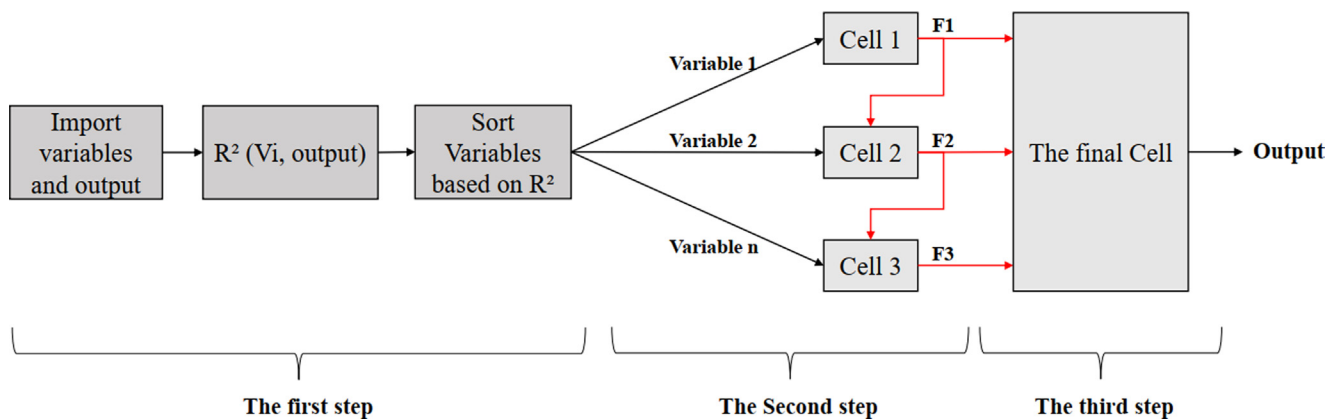
where the superimposed 'T' refers to the transpose.

3.2.2. Gene Expression Programming (GEP)

The Genetic Algorithm (GA) was first introduced by Holland [40], which was inspired by the natural biological evolution of living species. Different operators such as reproduction, selection,

crossover, and mutation inspired from natural biological evolution are integrated in the GA, and the solution as an individual is signified in fixed chromosome form. Similarly, Genetic Programming (GP) as an extension of GA was first proposed by Cramer [41]. GP requires a fitness function to evolve a population of individuals by imitating the basic principles of Darwinian evolution. It generates new solutions using an iterative process involving the probabilistic selection of the fittest solutions and their variation by means of crossover and mutation [42]. GEP [43] is an extension of the conventional GP and uses populations of individuals, selects them according to their fitness, and introduces a genetic variation using genetic operators. The most important difference between the three algorithms lies in the nature of the individuals: in GAs, the individuals are symbolic strings of fixed length, the so-called chromosomes whereas in GP, non-linear entities of different sizes and shapes form the individuals. They are called functions or parse trees. And finally, in GEP, the individuals, called Expression Trees (ET), are non-linear entities of different sizes and shapes, which are encoded as linear strings of fixed length (the genome or chromosomes) [44]. The GEP process begins with the random generation of the chromosomes of the initial population. Then, the chromosomes are expressed, and the fitness of each individual is evaluated. The individuals are selected according to the fitness, leaving progeny with new traits. The individuals of this new generation are subjected to the same process: expression of the genomes, confrontation of the selection environment and reproduction with modification. The process is repeated for a certain number of generations or until a solution has been found.

GEP is a strong tool to predict various phenomena and material properties [45]. In this study, we utilize the GEP model to predict the compressive strength of concrete. This model is created by using a set of different functions and terminals. A set of different functions, which are used in this study, includes the basic functions, inverse trigonometric function, trigonometric function, exponential function, hyperbolic function, user-defined functions, and a combination of them. A set of terminals consists of input variables, their combination, and constant values. In this approach, a random population of data is used by GA to find a suitable function to predict the output, which is the compressive strength of concrete in this study. GA selects the suitable input variables and functions between all functions and input variables to estimate the output. Several operators, named genes, are used for performing Genetic variations. Based on the GEP algorithm, the Roulette wheel selects data. Several genes reproduce the data. In the duplication process, the genes avoid unsuitable data and transfer suitable data to create the next generation. The main target of the mutation operator is related to the internal random optimization of the given chromosomes. Fig. 4 explains the process of GEP.

**Fig. 3.** The general form of SBSR.

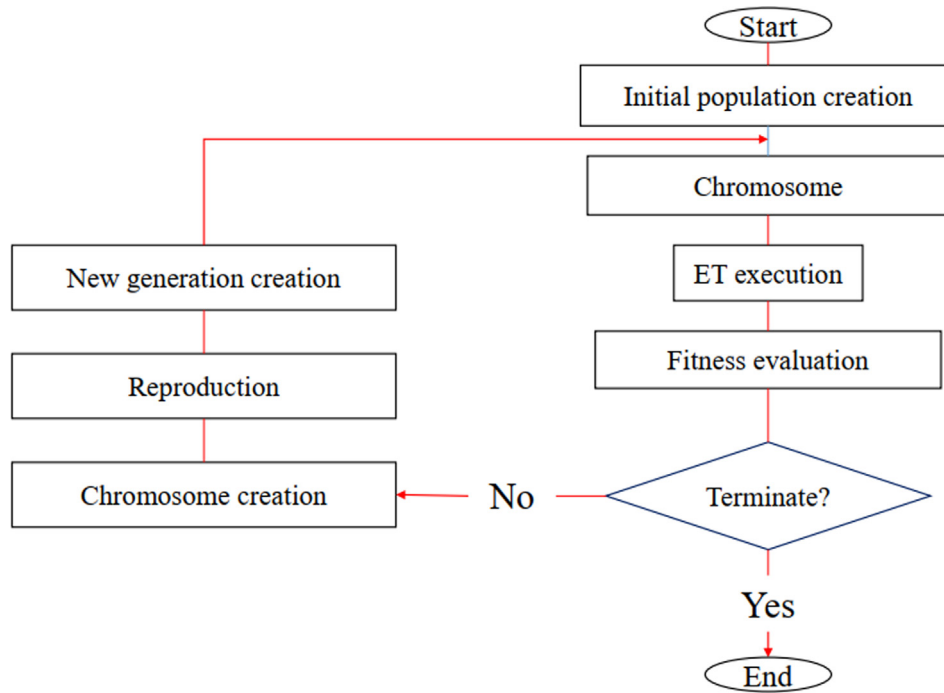


Fig. 4. The process of GEP.

3.2.3. Adaptive neuro-fuzzy inference system (ANFIS)

The fuzzy logic theorem was first proposed by Zadeh [46] to describe complicated systems. Recently, fuzzy logic methods have been used successfully in many applications [47,48]. The main problem with fuzzy logic is that there is no systematic procedure for the design of a fuzzy controller [49]. Indeed, the performance and accuracy of the fuzzy systems are related to the used rule-based and the parameters such as the Membership Functions (MFs).

An ANN has the ability to learn from the environment by input-output data sets, self-organizing its structure, and adapting it in an interactive manner [50]. ANFIS was proposed to capture the benefits of both Fuzzy Inference Systems (FISs) and neural networks in a single framework [51]. The ANFIS is a multi-layer feed-forward network, which uses neural network learning algorithms and fuzzy reasoning to map an input space to an output space. It has the ability to combine the verbal power of a fuzzy system with the numeric power of a neural system adaptive network [52,53]. In other words, ANFIS utilizes a linguistic model containing a set of If-Then fuzzy rules and the fuzzy sets to present the human-like reasoning style of a fuzzy system [54].

Two types of FIS are commonly used: Sugeno fuzzy and Mamdani models. Their defuzzification and aggregation approaches are different; therefore, the results of fuzzy rules are various for these systems. The number of rules grows with the number of premise-part variables in the Mamdani-type FIS. By growing the number of rules, the activity of assembling rules maybe complicate the relationships between the consequences and premises [55,56]. The Sugeno-type approach (or Takagi-Sugeno-Kang) is based on fuzzy inputs and a crisp output (linear combination of the inputs) that computationally is suitable and efficient. A systematic method is developed using the Sugeno approach to generate fuzzy rules from a given input-output data set. The sugeno method changes the sequence with an equation (function) of the input parameters. A weighted average is used by the Sugeno-type FIS to calculate the crisp output, although the technique of defuzzification of a fuzzy output is used by the Mamdani-type FIS. The first two parts of

the fuzzy inference process, including fuzzifying the inputs and applying the fuzzy operator, are identical; however, the Sugeno output membership functions are either linear or constant [57].

Fig. 5(a) demonstrates the general architecture of the ANFIS model, which includes two inputs and one output. Fig. 5(b) shows the fuzzy reasoning mechanism.

In the present investigation, a Sugeno FIS is used, and consequently, two Fuzzy If-Then rules are defined as a linear combination of input parameters, and the final output equals the weighted mean of outputs for each rule. A Sugeno FIS, which includes an output (f), and two input parameters (y and x), is represented as follows [58,59]:

$$\text{Rule 1 : if } x \text{ is } A_1 \text{ and } y \text{ is } B_1; \text{ then } f_1 = p_1x + q_1y + r_1 \quad (6)$$

$$\text{Rule 2 : if } x \text{ is } A_2 \text{ and } y \text{ is } B_2; \text{ then } f_2 = p_2x + q_2y + r_2 \quad (7)$$

where B_1 , B_2 , A_1 , and A_2 are the labels presenting the MFs for the inputs x and y , respectively. r_i , q_i , and p_i designate the Output Membership Function Parameters (OMFPs), which are the consequent parameters of i th rule.

As can be seen in Fig. (5), the architecture of a typical ANFIS consists of five layers. Each layers' functions are explained as follows:

The first Layer: Each node (i) in the first layer is a square node with a node function:

$$O_A^i = \mu_{A_i}(x) \quad (8)$$

where O denotes the output of each node in this layer; x and A_i are the input variable to node (i) and the label to present the MFs for the input, respectively. The Gaussian Membership Functions (GMFs), used here are given by

$$\mu_{A_i}(ln_1, \sigma_i, C_i) = \exp\left(-\frac{(ln_1 - C_i)^2}{2\sigma_i^2}\right), i = 1, 2 \quad (9)$$

$$\mu_{B_j}(ln_2, \sigma_j, C_j) = \exp\left(-\frac{(ln_2 - C_j)^2}{2\sigma_j^2}\right), j = 1, 2 \quad (10)$$

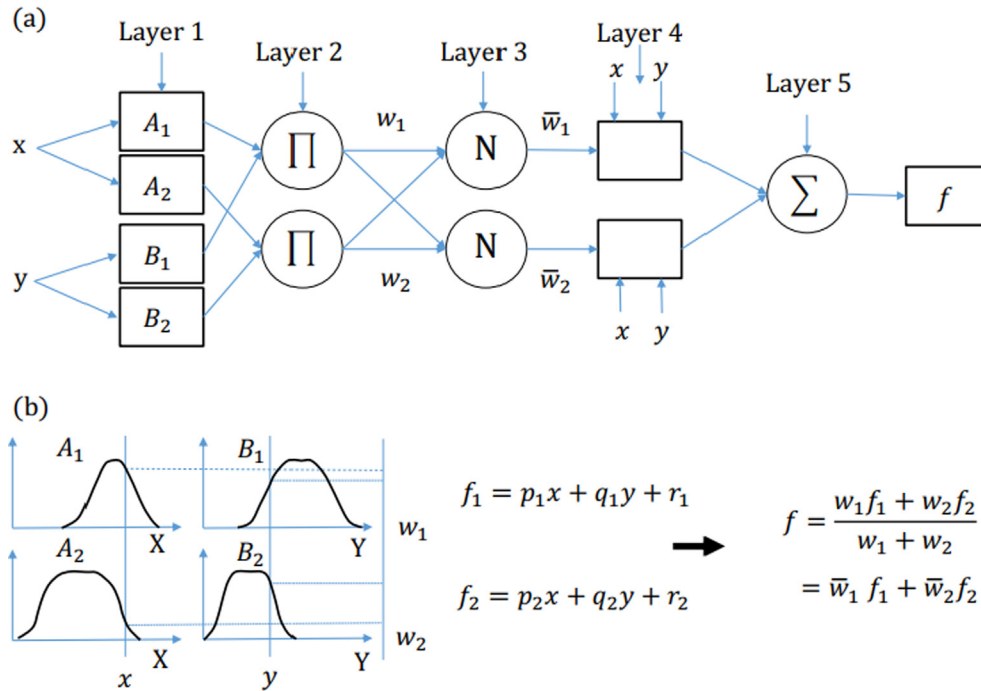


Fig. 5. The structure of ANFIS (a) The general architecture of ANFIS (b) Fuzzy reasoning scheme of ANFIS [46]

$\{\sigma_i, C_i\}$ and $\{\sigma_j, C_j\}$ being the premise parameters, which are governed form GMFs.

The second layer: Each node in the second layer is a circle node, which is labeled as N that multiplies the output signals, and their outputs are calculated as follows:

$$W_i = \mu_{A_i}(x) \mu_{B_i}(y) \quad i = 1, 2 \quad (11)$$

In this layer, the outputs of the nodes show the firing weight of the rules.

The third layer: Each node in the third layer is defined as a circle node, which is labeled as N that plays a normalization role. The i th node calculates the ratio of the firing weight of the i th rule to the summation of the firing weights of all rules as:

$$\bar{W}_i = \frac{W_i}{W_1 + W_2} \quad i = 1, 2 \quad (12)$$

The fourth layer: Each node in the fourth layer is a square node with a node function. Each output of this layer is determined as:

$$O_i^4 = \bar{W}_i f_i = \bar{W}_i (p_i x + q_i y + r_i) \quad i = 1, 2 \quad (13)$$

The parameters of layer 4 are related to as consequent parameters.

The fifth layer: There is one circle node in layer 5, which is a fixed node labeled by Σ , and defined as the signal node. This node computes the overall output, which is obtained from the summation of all incoming signals as

$$O_i^5 = \sum \bar{W}_i f_i = \frac{\sum W_i f_i}{\sum W_i} \quad (14)$$

There are two approaches in ANFIS. The first is a back propagation for all parameters. The second is a hybrid approach, which includes a least-squares estimation for the parameters associated with the OMFPs and the back propagation for the parameters associated with the input MFs [58].

3.2.4. High Correlated variables Creator Machine (HCVCVM)

HCVCVM is a new algorithm to create the new variables from the initial variables improving the performance of model. There are three steps in this method. In the first step, the new variables are created using several mathematical, trigonometric functions, their combinations, and user-defined functions. In the second step, the coefficient of determination between new variables and outputs is determined. In the second step, the coefficient of determination between new variables and the outputs is determined. The values of the coefficient of determination (R^2) between the initial input variables and the output are compared to the R^2 values between the new variables and the output. Among all new variables, only the variables, which have larger R^2 values with the output compared to the R^2 values of the initial variables with the output, are imported to the third step. In the third step, among all selected new variables in the second step, only the variables, which R^2 values with each other lower than the R^2 values between the initial variables, are imported to prediction models. On the other hand, HCVCVM tries to generate and find several new variables, which have more correlations with the output in comparison to initial variables but have lower correlations with each other. Therefore, they can independently improve prediction models. HCVCVM can contain several generations. Fig. 6, shows the process of the first generation in HCVCVM. Only new variables, which are selected in the first generation, are imported as input variables in our hybrid models. In the second generation, new variables can be created using the new selected variables in generation 1.

3.2.5. Using HCVCVM in prediction models

HCVCVM is used before SBSR, GEP, and ANFIS to create the new variables, which finally improves the accuracy of the models. First, better variables are specified based on HCVCVM. Then, these variables are imported to SBSR, GEP, and ANFIS. Based on the results of HCVCVM, the coefficient of determination (R^2) value of 6 new variables created from UPV will be higher compared to the R^2 value of UPV. Moreover, the R^2 value of the new RN will be higher than the old R^2 value. The functions, which are used to create these new variables from UPV and RN, are shown in Table 3.

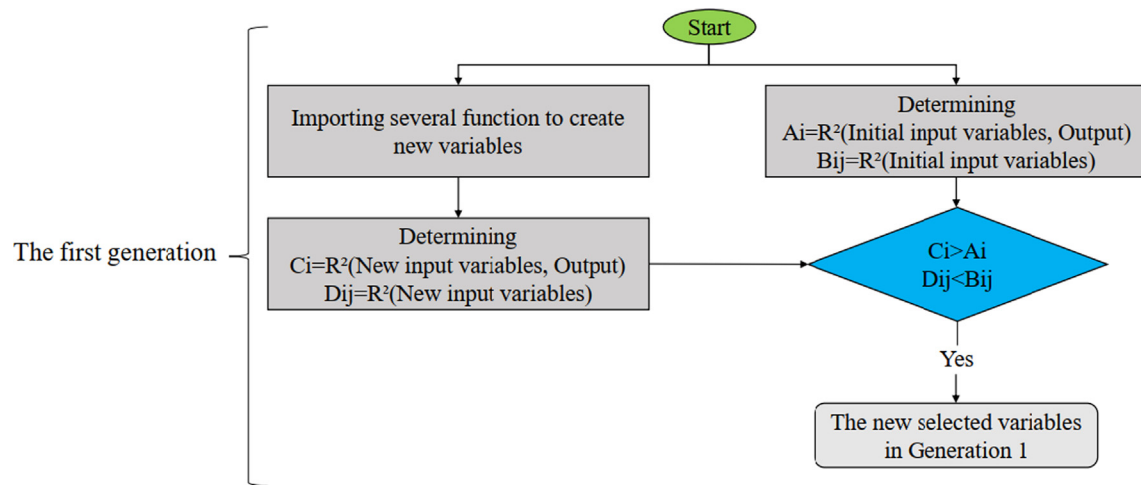


Fig. 6. The process in HCVCML.

Table 3

The coefficient of determination (R^2) between new variables and output.

New variables	UPV2	UPV5	UPV6	UPV7	UPV8	UPV9	RN5
R^2 (new variable, output)	$\cos(a)$	a^2	a^3	a^4	a^5	$\exp(a)$	b^2
	0.454	0.455	0.464	0.471	0.476	0.478	0.776

As mentioned in section 3.2.4, not only the coefficient of determination between the new variables and the output should be higher than the coefficient of determination between the initial variable and the output, but also the coefficient of determination between the new variables should be lower than the coefficient of determination between the initial variables. Therefore, as shown in Tables 3 and 4, the new variables are selected, which pass these two rules.

4. Results

As mentioned before, 70% of the dataset was used to train the models; then, the model was utilized to predict the remaining. The same data was used to train all models. Table 5 summarizes the statistical parameters of six models in the training and testing dataset. ANFIS performs better than all the other single models with respect to all statistical parameters in the

training dataset, and SBSR is better than GEP according to the coefficient of determination, RMSE, and NMSE. In contrast, GEP performs better than the other single models in the testing dataset. Although the coefficient of determination of SBSR is better than those of ANFIS in the testing dataset, ANFIS shows better results with respect to RMSE and NMSE. The results show that the best single model based on statistical parameters cannot be selected.

As Table 5 demonstrates, HCVCML positively affects the results of SBSR and ANFIS because not only the R^2 value increases – compared to the associated single models, but also the RMSE and NMSE are less. HCVCML-GEP performs better than GEP in the training dataset, although, in the testing dataset, GEP yields better results in the RMSE and coefficient of determination.

Table 6 lists the error terms, including the maximum positive error, maximum negative error, and MAPE. ANFIS is the best single model. HCVCML-ANFIS performs better than ANFIS with respect to

Table 4

The coefficient of determination (R^2) between new variables.

R^2 (new variables)	(UPV2, RN5)	(UPV4, RN5)	(UPV5, RN5)	(UPV6, RN5)	(UPV7, RN5)	(UPV8, RN5)
R^2	0.484	0.434	0.481	0.493	0.502	0.509

Table 5

The results of the statistical parameters for all models.

Models	R^2	RMSE	NMSE	Fractional bias	R^2	RMSE	NMSE	Fractional bias
SBSR	0.761	46.813	0.084	$-1.56E-05$	0.935	51.119	0.189	$7.97E-05$
HCVCML-SBSR	0.788	43.809	0.100	$-1.41E-05$	0.926	43.538	0.130	$7.20E-05$
GEP	0.756	47.824	0.105	$-2.54E-05$	0.943	36.979	0.109	$7.22E-05$
HCVCML-GEP	0.767	46.207	0.102	$-1.41E-05$	0.911	39.273	0.108	$4.36E-05$
ANFIS	0.824	39.901	0.090	$-1.66E-05$	0.930	39.297	0.110	$8.24E-05$
HCVCML-ANFIS	0.862	35.742	0.064	$-1.35E-05$	0.950	32.628	0.084	$8.03E-05$

Table 6

The results of the error terms for all models.

Error terms	SBSR	HCVCML-SBSR	GEP	HCVCML-GEP	ANFIS	HCVCML-ANFIS
The maximum positive error	102.3	31.732	35.813	39.175	28.842	31.706
The maximum negative error	-89.7	-81.815	-86.275	-86.320	-85.303	-79.363
MAPE	15.73	13.932	13.658	12.856	12.174	10.145

the error term. It also leads to better results compared to HCVCMSBSR and HCVCMEP. The maximum positive and negative errors of HCVCMSBSR are less than those of HCVCMEP, but MAPE of

HCVCMEP is less than those of HCVCMSBSR. The results show that HCVCMEP is more useful when this method is used with SBSR and ANFIS.

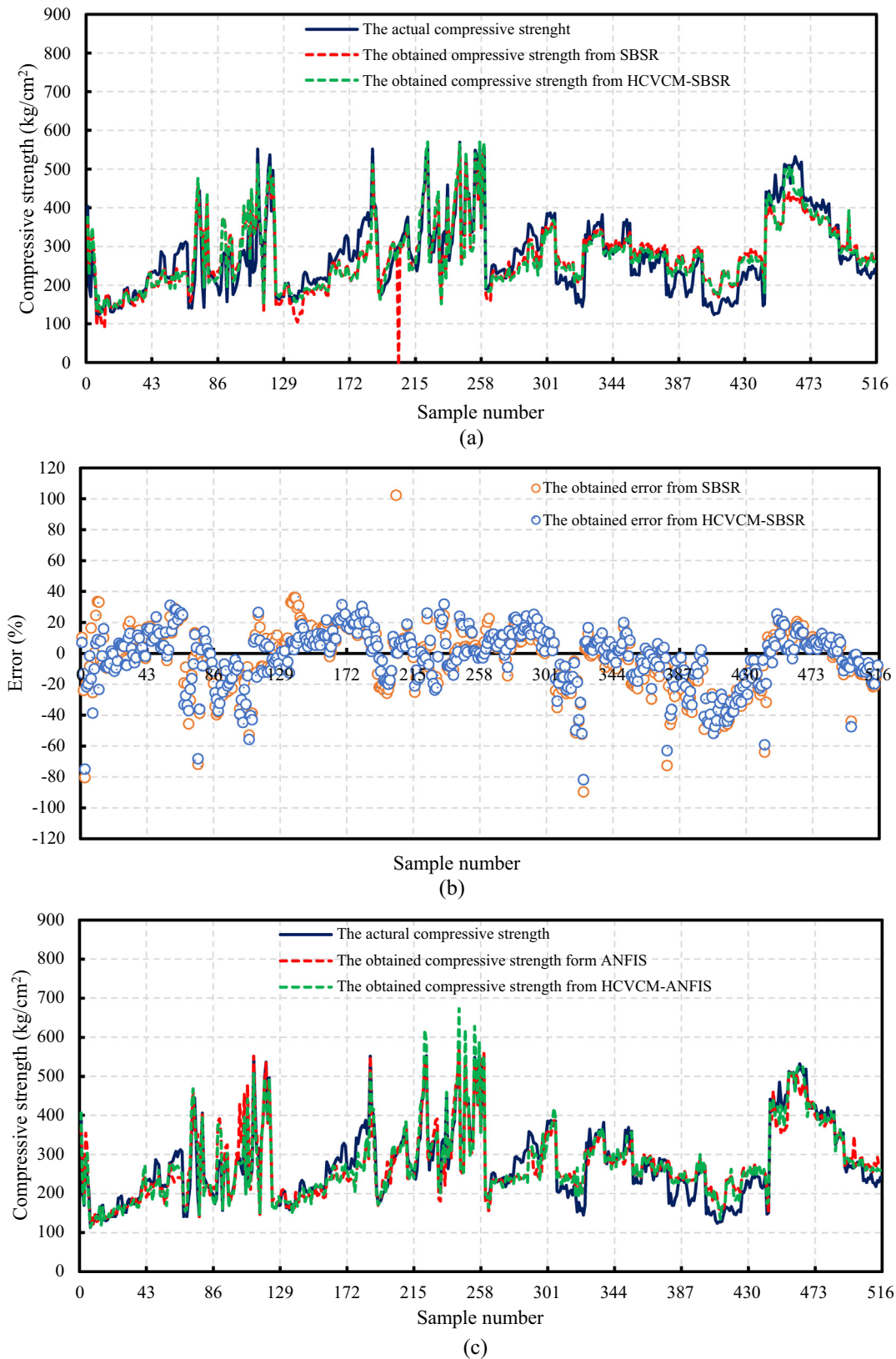
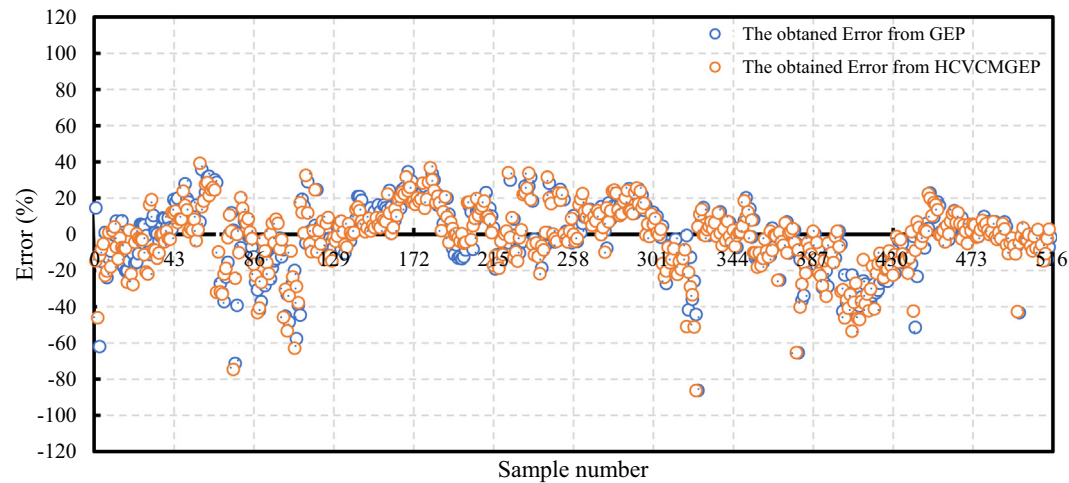
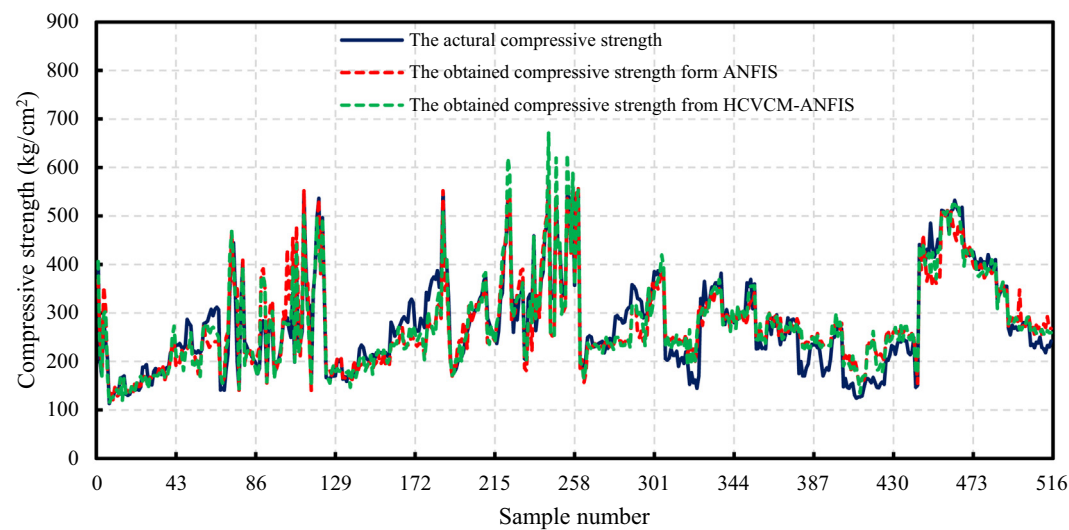


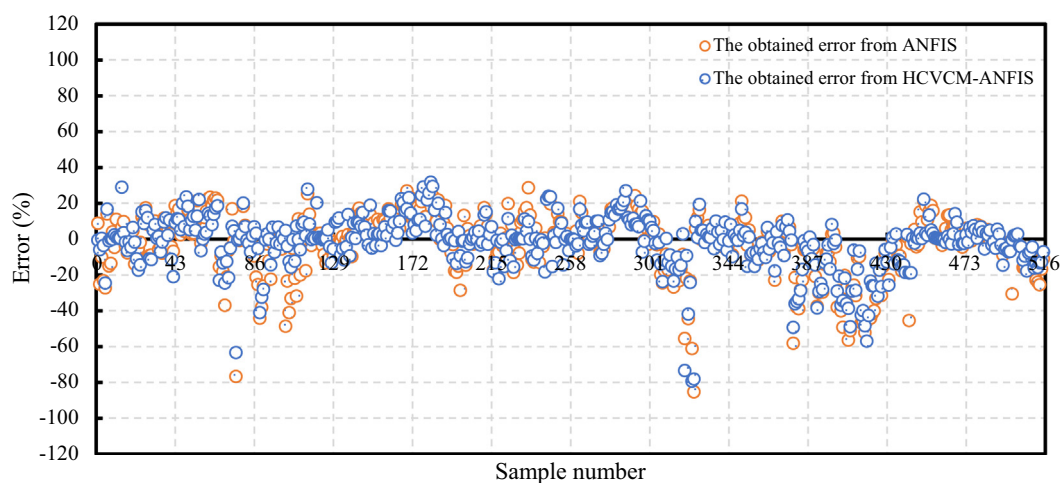
Fig. 7. The actual and predicted compressive strength of concrete and obtained errors from models. (a) The predicted compressive strength by SBSR and HCVCMSBSR. (b) The obtained error from SBSR and HCVCMSBSR. (c) The predicted compressive strength by ANFIS and HCVCMEP. (d) The obtained error from ANFIS and HCVCMEP. (e) The predicted compressive strength by ANFIS and HCVCMEP. (f) The obtained error from ANFIS and HCVCMEP.



(d)



(e)



(f)

Fig. 7 (continued)

Fig. 7(a) shows the comparison of the actual compressive strength and the predicted values using SBSR and HCVCMEP-SBSR. The HCVCMEP-SBSR results better predict the actual compressive strength. Fig. 7(b) depicts the error of these models. HCVCMEP

decreases the error in SBSR. Figs. 8(c) and (d) compare the predicted compressive strength by GEP and HCVCMEP-GEP with the actual compressive strength. HCVCMEP is obviously not useful for GEP because GEP was not used the new variables, which were pro-

Table 7

The computational time for all models.

Model	SBSR	HCVCM-SBSR	GEP	HCVCM-GEP	ANFIS	HCVCM-ANFIS
Computational time (sec)	1	4	1432	1427	4	759

posed by HCVCM. On the other hand, GEP changed all new variables, which were created by HCVCM; therefore, the results of HCVCM-GEP cannot improve the results of GEP. Fig. 7(e) shows the predicted compressive strength using ANFIS and HCVCM-ANFIS, whereas Fig. 7(f) illustrates the obtained errors from ANFIS and HCVCM-ANFIS.

Table 7 shows the computational time for all models to predict the compressive strength of concrete. SBSR includes several regression techniques; thus, the computational time of this model is lower compared to the other models. The computational cost for GEP and HCVCM-GEP is nearly identical as we stopped both GEP and HCVCM-GEP in the same generation. The computational time for ANFIS is only 4 sec because there are only two variables. There are more than 7 variables in HCVCM-ANFIS which significantly increases the computational cost.

5. Conclusion

In this study, three single models, including SBSR, GEP, and ANFIS, and three hybrid models, including HCVCM-SBSR, HCVCM-GEP, and HCVCM-ANFIS, were used to predict the compressive strength of 516 concrete samples with various UPV and RN. In the following, the main findings are summarized:

- 1- HCVCM aims to produce better variables with respect to the coefficient of determination between the new variables and the output. On the other hand, the coefficient of determination between new input variables and outputs is more than those between the initial variables and output.
- 2- Selecting the new variables with lowest correlation to each other and highest correlation with the output can improve the accuracy of the models.
- 3- HCVCM improves the accuracy of SBSR and ANFIS to predict the compressive strength of concrete.
- 4- HCVCM is an adaptive method for GEP in predicting the compressive strength of concrete.
- 5- HCVCM improve the accuracy SBSR 2.6% in coefficient of determination, 6.5% in RMSE, 3.75% in NMSE, 11.4% in MAPE, 69.2% in maximum positive error, and 8.9% in the maximum negative error.
- 6- HCVCM-ANFIS is selected as the best model to predict the compressive strength of concrete, HCVCM improve the accuracy ANFIS 5% in coefficient of determination, 10% in RMSE, 3% in NMSE, 20% in MAPE, and 7% in the maximum negative error.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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